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Pioneering 25-Stages Tool-Free Fishbone Fracture Acidizing Stimulation in Openhole Horizontal Well: Field Implementation in Low Permeability Carbonate Reservoir in Central Iraq

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Abstract

The irregular wellbore trajectory of open hole horizontal wells poses significant challenges to tool accessibility, especially for efficient and low-cost multistage stimulation. In this paper we will tackle these challenges by sharing our experience in combining tool-free fishbone-like fracture acidizing stimulation and detailing the methodology and performance in unlocking the potential of our reservoir with reduced capital expenditure (CAPEX).

The target Sa'adi Formation in Central Iraqi exhibits complex lithology with wackestone, packstone, and chert, where vertical well acid-fracturing achieved only 83 bbl/d. The first open hole horizontal well, S-1H, was drilled in Nov. 2024, and maximize reservoir potential without secondary completion is the biggest challenge for stimulation. The solution deployed 2 $\frac{3}{8}$ -in coiled tubing with quad-nozzle jetting tools, creating self-propped fractures via 1.0 m³/min sand slurry (180 m/s velocity, 2-3m radial penetration). Each stage combined: Progressive sand volumes (0.5 to 1.0 m³) with real-time drag monitoring to prevent sticking; 20% HCL injection (20-50 m³/stage) at maximum pump rates @ 63MPa; Centralizer-assisted CT movement. Sand jetting first established infinite-conductivity entry points (total 11 m³ quartz sand), followed by acidizing (total 727 m³ HCL) to extend dissolution beyond near-wellbore regions. Operational parameters were dynamically optimized using pressure-decline analysis between stages, achieving full 1,200m lateral coverage without mechanical isolation.

Post-treatment well testing yielded 10,647 bbl/d—127 $\times$ vertical cased-hole benchmarks. Diagnostic data confirmed: 25 distinct stimulation zones (Fiber optic logging), 2.2m average fracture penetration (Pressure fitting), and wormhole density increase from 3/cm² to 28/cm². Fiber optic logging revealed that 23 primary fractures with high density wormholes each, creating a fishbone-like infinite conductivity zone. The operation achieved 92% success rate despite severe open hole condition, completing all stages within 39 hours—70% faster than conventional ECP methods. Total costs were reduced by \$2.0 MM versus cased-hole completion, with zero CT stuck incidents. Sustained production over 90 days demonstrated stable fracture conductivity, outperforming offset wells' 40% decline rates.

This strategy delivers the industry's first tool-free fishbone fracture stimulation technology via sand jetting & acidizing in open hole wells. It provides a technical solution for the low-permeability and complex lithology carbonate reservoirs in the Middle East.

Introduction

The low permeability carbonate reservoir in central Iraq is within the Mesopotamian Fore deep Basin, located northeastward to the Zagros Fold Belt and westward to the Widian Basin of the Interior Platform (Xiao, et al., 2023). The first well of this low permeability carbonate reservoir was drilled in 1975 and a total 17 wells drilled before ZHO took over, and four layers of the carbonate reservoir were confirmed: Hartha, Sa'adi, Tanuma, and Khasib. Since the official development of the oil field, open hole horizontal well with acidizing completion has been the main development strategy.

Petrophysical Characteristics

Core description and geological logging analysis reveal that these four low-permeability carbonate reservoirs are buried at depths of 1,800 to 2,000 meters, with normal pressure and temperature gradients, an average porosity of 20%, and an average permeability of 1.73 mD. The lithology primarily consists of light gray to gray packstone, grainstone, and wackestone with abundant bio-fragments such as foraminifers, algae and shells, etc. The reservoirs spaces are dominated by vugs, interparticle pores, dissolution and moldic pores. Overall, the pore connectivity is relatively poor, as shown in Fig. 1.

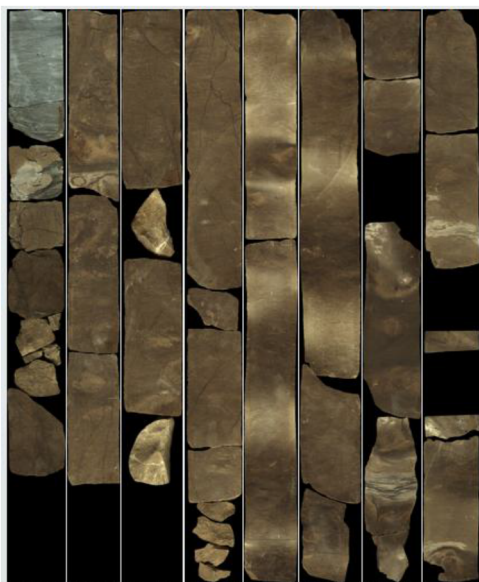


Figure 1—Core samples of the low permeability carbonate reservoir

Systematic petrophysical tests shows that the carbonate reservoirs can be classified into two categories: Medium porosity and low permeability in H/T/K reservoirs, and the low porosity and low permeability in Sa'adi reservoir. The pore-throat characteristics of the Sa'adi reservoir, including pore-throat radius, sorting, and connectivity, are all inferior to those of the other three formations. Furthermore, the Sa'adi reservoir is an oil-bearing muddy carbonate reservoir containing packstone and wackestone. The presence of insoluble minerals within this reservoir adversely affects the performance of conventional acidizing treatment, requiring the use of unique acidizing technology and acid formulation. Therefore, operators will face huge challenges in the economic balance of completion, stimulation, and capital expenditures (CAPEX).

Mechanism of Conventional Acidizing

Since 1895, acid stimulation techniques represented by acid wash, matrix acidizing and acid fracturing has been employed to stimulate low-permeability carbonate oil and gas reservoirs (Bryan et al., 2020). Matrix acidizing currently stands as the predominant stimulation method for such reservoirs. Its principle involves injecting acid at low rates to remove the comprehensive skin within the range of about 0.5-2.0m near the wellbore, reducing the flowing pressure differential and recovering productivity. A critical requirement of this operation is maintaining injection pressure below the formation breakdown pressure, and the volume of injected acid depends on the severity of formation damage. Matrix acidizing offers advantages including feasible operation, lower cost, and excellent effectiveness on severely damaged wells (Gou et al., 2024).

Conversely, for tight carbonate formations characterized by low porosity and low permeability, the primary limitations on production are poor connectivity between the reservoir and the wellbore, coupled with a limited drainage area. These reservoirs necessitate more aggressive acid treatments, for which acid fracturing is recommended. This technique is a combination of hydraulic fracturing and matrix acidizing. It involves first creating tens to hundreds of meters long artificial fractures to deep reservoir from the wellbore. Subsequently, soluble acid is pumped to form self-supporting high-conductivity channels using uneven etching of the fracture wall. While acid fracturing requires high-pressure pumping, complex equipment and operation and high costs, it remains the preferred stimulation technology for tight carbonate reservoirs (Guo et al., 2021).

For open hole horizontal wells, due to the limitations of irregular open-hole wellbore walls, lack of mechanical segmentation without casing, and small-sized wellbore, etc., coiled tubing equipped with downhole tools is typically employed, performing stage-by-stage acidizing while being pulled up from the bottom, as shown in Fig. 2. The primary objective of this treatment is to remove drilling-induced skin damage. The effective invasion ranges between 0.5 and 2.0 meters. The acid creates wormholes that connect the wellbore to the reservoir, restoring the well to its original productivity potential. It's important to note that this approach generally does not achieve significant production enhancement above the original potential.

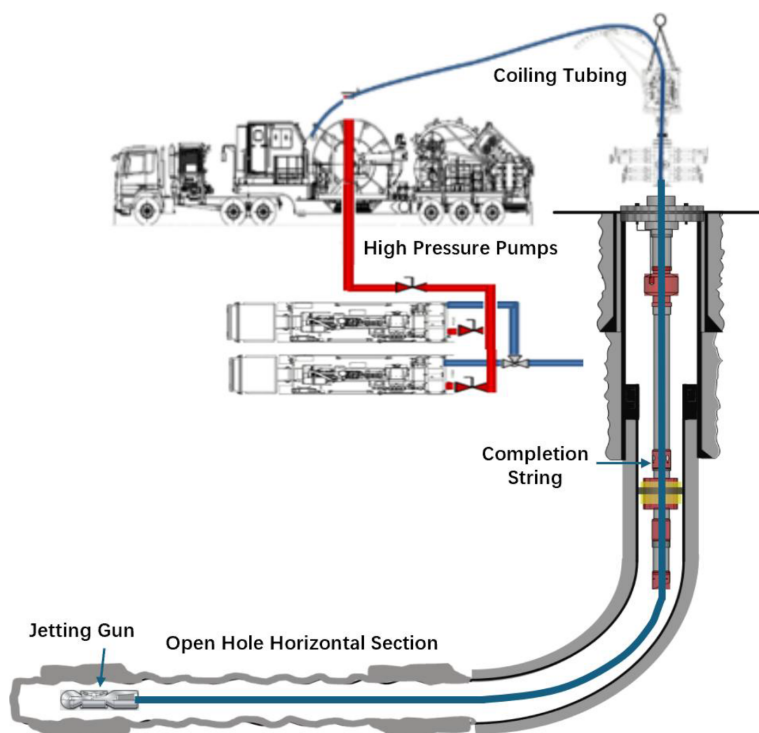


Figure 2—Conventional Open hole horizontal well acidizing within coiled tubing

This technique is carried out after the horizontal well is completed. Coiled tubing is run through the completion string to the bottom of the well. Critically, this operation eliminates the need for a workover rig to trip the production tubing, resulting in significantly lower operational costs and a shorter job duration. Furthermore, coiled tubing deployment readily supports repeated acid stimulation treatments. Therefore, coiled tubing acidizing from toe to heel has become the main stimulation method for open hole horizontal wells in low-permeability carbonate reservoirs in central Iraq, but the performance varies for different underground conditions.

Evaluation and Performance of Acid Stimulation

Since 2019, the oilfield has entered a period of formal, large-scale development and construction, and gradually started conventional stimulation such as the first horizontal well acidizing and the first acid fracturing. Since then, the oilfield has ushered in a new era of rapid development. Reservoir stimulation represented by acidizing and acid fracturing has been identified as the primary strategy for enhancing production. By the end of 2024, nearly 100 acid stimulation treatments (including both matrix acidizing and acid fracturing) had been completed throughout the entire oilfield, of which more than 65% are horizontal wells. The field implementation encompassed a diverse range of acidizing technologies, various acid systems and different scales. Concurrently, driven by ongoing research and the accumulation of field experience in recent years, acid stimulation technologies have undergone continuous innovation. Notable achievements include: The successful implementation of acidizing in a 2500-meter open-hole well using 2-inch coiled tubing, re-entry acidizing operations in dual-lateral wells, and 25-Stages tool-free fishbone fracture acidizing stimulation in open hole horizontal well.

Acidizing in Medium Porosity Low Permeability Carbonate Reservoir

Matrix acidizing in horizontal wells is performed to remove potential drilling-induced formation damage and to generate wormholes that connect the wellbore to deeper reservoir zones, thereby restoring production (Guo et al., 2014). The acidizing method for medium-porosity and low-permeability carbonate reservoirs in this oil fields is to use 2" and 2-3/8" CT to run to the bottom of the well, and then pull back for pinpoint acidizing. The bottom hole assembly (BHA) is CT + Roll-on connector + Downhole Flow Control Valve (DFCV)+ Rigid centralizer + 6*4mm nozzles. DFCV is the key component for emergency handling, such as well blowout in case of CT collapse. In order to ensure the successful running and reduce the potential danger during RIH and POOH, the coiled tubing speed is designed to be less than 25 m/min, and the axial force analysis is carried out. The 2" coiled tubing load capacity is 30639kg and the maximum injector pulling capacity is 45663kg, effective axial force during tripping in: Maximum 7963kg@2045.7m and Effective axial force during tripping out: Maximum 13445kg@3873m.

The acid system primarily composed of 20% hydrochloric acid (HCL), with additives including corrosion inhibitors, iron stabilizers and multi-functional agents that reduce surface tension and have good demulsification effects. The acid intensity ranges between 0.3 m³/m and 0.7 m³/m, with total acid volume varying from 300 m³ to 1000 m³, which is optimized according to the geological characteristics of the reservoir. Significantly, starting from 2024, to further increase acid pumping capacity, 2-3/8" coiled tubing was implemented for the first time. The upgrade enabled pumping rate to be increased from 0.3, 0.7 to the current 1.1 m³/min, enabling effective deep penetration acidizing, as shown in Fig. 3.

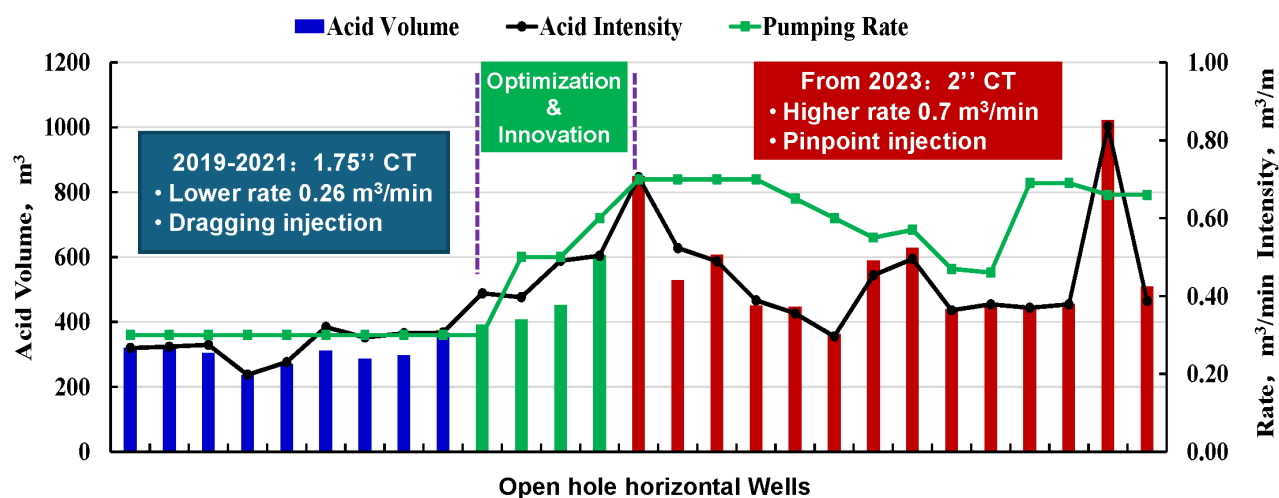


Figure 3—Improvement of acidizing parameters

Concurrently, engineers re-optimized seven key operational parameters: acid concentration, injection rate, pump pressure, coiled tubing diameter, acid displacement method, acidizing intensity, and nozzle size. This effort led to the establishment of pinpoint acidizing techniques. The acid-etched wormhole after acidizing increased from 0.8m to 2.0m, the highest test production after acidizing in vertical wells exceeded 10,000 bbl/d, with maximum production increases exceeding 11 times, and the average test production of horizontal wells reached 2268bbl/d @ 32/64". This series of achievements provides valuable experience for the future technical research and development of multi-stages stimulation technologies.

Acid Fracturing in Tight Muddy Carbonate Reservoir

Regarding acid fracturing, the technologies currently implemented in the oilfield include: two-stage alternating acid fracturing, and closed fracture acidizing. The objective is to utilize high pressure to create artificial fractures within the reservoir. Subsequent acid then heterogeneously etches the fracture wall, forming self-propping and highly conductivity flow channel (Xiao et al., 2020). The downhole tool assembly for acid fracturing consists of an integrated 3-1/2" tubing + Tubing Conveyed Perforating (TCP) string, which enables one trip into the well conducting perforating, acid fracturing and testing. Surface equipment primarily includes: 4 sets of 2000 HP fracturing pump trucks, fluid storage tanks and high/low-pressure manifolds, and the operation pressure limit is 10,000 psi.

To meet distinct objectives at different stages of the acid stimulation, a variety of liquid systems are recommended. 15% diverting acid, which achieves the purpose of plugging and diversion by continuously increasing viscosity to block flow paths; 20% gelled acid, which reduces the migration rate of hydrogen ions, thereby slowing down the acid-rock reaction rate, enabling deeper acid penetration. The pad fluid is a low-friction, low-residue, low-fluid-loss guar-based linear gel, which provides a friction reduction rate reaching 80%, facilitating high-rate pumping. 15% HCL is applied for closed fracture acidizing to rapidly dissolve the fracture entry (near-wellbore region), enhancing conductivity. But the pumping pressure must be maintained below the fracture closure pressure. Water is used for displacement after the fracturing and for flushing.

As a representative of low porosity and low permeability, the Sa'adi reservoir, the oil field conducted acid fracturing tests on two wells in 2019, both with unsatisfactory performance. Taking EB-X well as an example, the operation involved a mini-frac test followed by two-stage alternating acid fracturing. A total of 94m³ of linear gel and 122 m³ of 20% gelled acid were pumped, with a maximum pumping rate of 3.7m³/min and a peak pumping pressure of 36.7MPa, as shown in Fig. 4. Post-frac production test from this well was only 101 bbl/d, and another vertical well in the same layer produced almost no oil after fracturing. These results indicate that this tight reservoir cannot achieve economical and efficient development using

conventional well type and stimulation method. Horizontal well must be considered and matched with appropriate reservoir stimulation technologies.

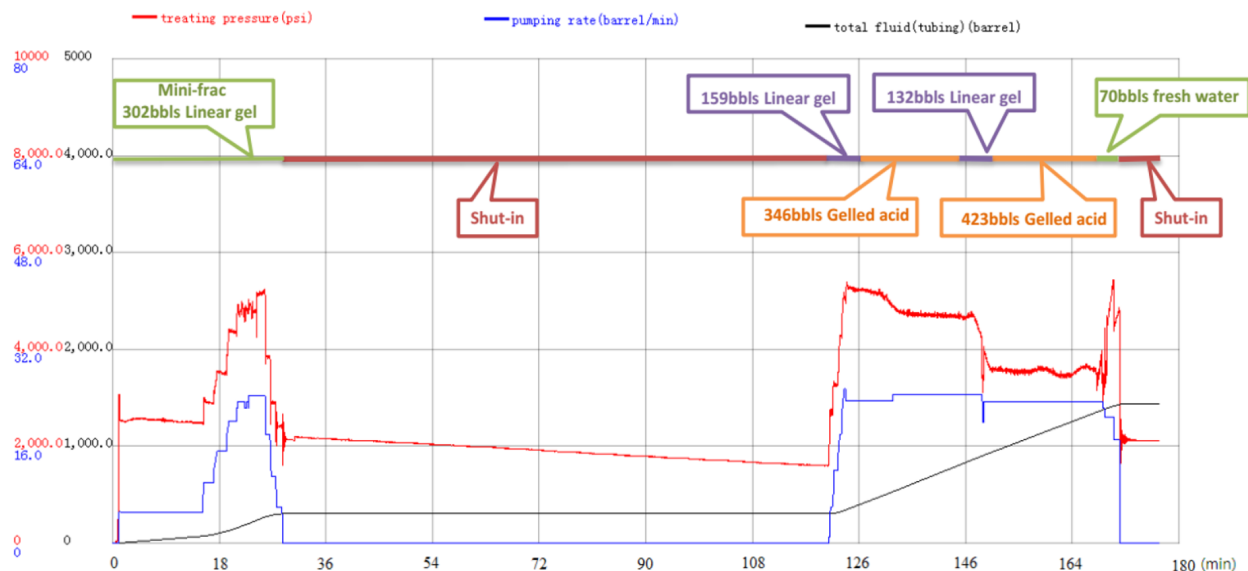


Figure 4—pumping curves of acid fracturing in EB-X well. The acid fracturing started with a mini-frac test utilizing linear gel and shut in for 90 minutes to monitor fracture closure. In the main stimulation, linear gel and gelled acid were alternately injected at sustained rates of 23.3–25.5 bpm @ 3,523–5,202 psi. Total fluid volumes reached 291 bbls of linear gel and 769 bbls of gelled acid. Shut in pressure is 2,150 psi.

Methodology of Tool-Free Fishbone Fracture Acidizing Stimulation

The Sa'adi formation is a low-porosity, low-permeability, tight, muddy carbonate reservoir with a temperature of 77°C and pressure coefficient of 1.16 @ TVD depths of 2200m. Conventional dragging matrix acidizing (pumping acid while pulling the CT back from toe to heel) for skin removing fails to achieve an economic production rate of at least 700 bbl/d in openhole horizontal wells. Standard coiled tubing acid fracturing can't maintain sufficient energy at downhole to breakdown the reservoir due to large friction losses, making it impossible to form a high-conductivity channel initiating tens to hundreds of meters into the formation. Consequently, the stimulation methodology must shift from damage removal/productivity restoration to constructing high-conductivity artificial fractures for production enhancement. Open-hole multistage stimulation faces inherent limitations: The irregular wellbore trajectory limits the entry of large size tools, mechanical isolation tools (e.g., bridge plug) are unsuitable due to the rough open-hole wellbore wall, and the exposed long open-hole section causes uncontrolled fluid intake preventing precise staged stimulation.

On the contrary, while casing recompletion with mechanical bridge plugs or perforated liners with external casing packers (ECPs) enables accurate stimulation, these approaches incur millions of dollars cost increments. To address these challenges and achieve precise, efficient, and cost-effective stimulation of tight carbonate reservoirs in open-hole horizontal wells, the following requirements must be met: Larger coiled tubing diameters to ensure high-rate fluid supply; simplified bottomhole assemblies that can pass through the 3-1/2" tubing; high-velocity jetting tools with mechanical integrity for multi-stage operations and sufficient rock-breaking energy; jet fluids with superior friction reduction maintaining adequate proppant concentration at low rates.

Based on the above requirements, the research team developed the Tool-Free Fishbone Fracture Acidizing Stimulation strategy, as shown in Fig. 5, integrating four core processes: High-velocity acid jetting for fracture initiation; Fluid-loss-controlling & soft-sealing for targeted zonal isolation; Abrasive sand jetting to create fractures and proppant supporting; Deep-penetration and wormhole propagation.

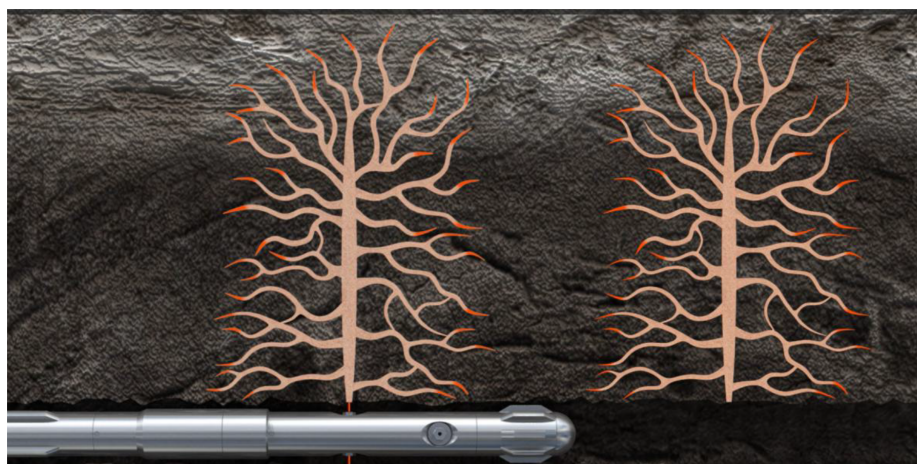


Figure 5—Schematic of Tool-Free Fishbone Fracture Acidizing Stimulation. Abrasive sand jetting creates large-diameter proppant-filled artificial fractures, deeply penetrating acid generates branching micro-fractures and wormholes, and ultimately forming fishbone-shaped high conductivity fracture networks.

Optimization

BHA

Engineers conducted a comparative analysis of friction between 2" (ID: 50.8mm) and 2-3/8" (ID: 60.3mm) coiled tubing. Surface water testing revealed that 5,000 meters of 2" coiled tubing experienced approximately 31.0 and 45.0 MPa of friction pressure loss along its length at flow rates of 0.28 m³/min and 0.34 m³/min, respectively. In contrast, the same-length of 2-3/8" CT exhibited maximum friction pressure loss of only 15.8 MPa under equivalent rates – confirming the significant impact of diameter variation on hydraulic resistance. However, when the pumping rate of 20% hydrochloric acid reaches 0.9 m³/min, which has no friction reduction effect, the friction reaches 71.5 MPa. Therefore, it is essential to select a fracturing fluid with a friction reduction rate of at least 75% as the sand slurry fluid, enabling sustainable rates exceeding 1.0 m³/min. 2-3/8" coiled tubing is recommended for operation.

Conventional perforating nozzles cannot meet high-velocity jetting/cutting requirements, necessitating the use of sandblasting perforating tools. Downhole tools include Shear Release Disconnect, Hydraulic Disconnect, Jetting Gun (4*4.56mm nozzles), Circulation Valve, Crossover, and Guide Tool. The Shear Release Disconnect and Hydraulic Disconnect are used to release the jetting gun and BHA in an emergency if the tool becomes stuck.

Jetting Parameters

The sand blasting perforation technology in open hole utilizes a surface pump system to pump a fluid mixed with a certain concentration of sand into a set of nozzles. The fluid is then accelerated through nozzles, forming a high-velocity slurry jet that impacts formation rock, creating perforated flow tunnels of a certain diameter and depth between the wellbore and the reservoir. The energy conversion can be described by Bernoulli's principle.

$$P_i + \frac{1}{2}\rho_f v_{fi}^2 = P_{i+1} + \frac{1}{2}\rho_f v_{fi+1}^2 + h$$

Where, P is pressure, ρ is density, and v is velocity. The subscript f represents fluid.

The fluid's pressure energy is converted into kinetic energy, generating a high-velocity abrasive jet. Within perforation tunnel, jet velocity rapidly decreases upon contact with the borehole wall, converting kinetic energy back to pressure. When the pressure exceeds the formation's fracture pressure, inducing fractures at the tunnel tip (Luan, et al., 2025). The abrasive jet's velocity reaches its highest value at the nozzle outlet, while its pressure reaches its lowest. Consequently, a "low-pressure zone" forms near the

nozzle outlet. This low-pressure zone facilitates tunnel erosion while functioning as a hydraulic barrier (soft-sealing effect), enabling tool-free isolation – the foundation for precise multi-stage stimulation.

The net force acting on the fluid and sand during acceleration within the nozzle is F :

$$\begin{cases} F_{vi} = \frac{1}{2} K_d \rho_f (v_{fi} - v_{si})^2 \\ F_{si} = \frac{1}{12} \pi d^3 \rho_f \left(\frac{2v_{fi}^2 \tan \theta}{r_i} - a_i \right) \\ F_{pi} = -\frac{1}{3} \pi d^3 \frac{\rho_f v_{fi}^2 \tan \theta}{r_i} \\ F_i = F_{vi} + F_{si} + F_{pi} \end{cases}$$

Where, F_v is viscous force, F_s is mass force, F_p is pressure gradient force, K_d is resistance coefficients, d is diameter, θ is the semi-cone angle of the nozzle's converging section, r is the radius, and a is the acceleration. The subscript s represents sand. When the sand accelerates into the cylindrical section, the nozzle's cross-sectional radius remains constant, $\theta = 0$, $\Delta P = 0$, and the sand is subject only to viscous forces. Therefore, the total net force is:

$$F_i = \frac{1}{2} K_d \rho_f (v_{fi} - v_{si})^2$$

The motion of sand within the nozzle can be described by the following equation:

$$\begin{cases} S_{j+1} = S_j + v_{sj} \times \Delta t \\ v_{sj+1} = v_{sj} + a_j \times \Delta t \\ a_j = \frac{6F_j}{\pi \rho_s d^3} \end{cases}$$

Where S is the distance within the nozzle. F_v is viscous force. Assuming a surface pump rate, the velocity field of the sand and fluid within the nozzle can be calculated using the above theory, as shown in Fig. 6. Therefore, with the velocity field as the optimization objective, the sensitivity analyses of the semi-cone angle (θ) and convergent section length can be used to design the critical parameters of the nozzle. Furthermore, for a defined nozzle structure, the surface injection rate can be optimized according to the reservoir physical properties to ensure that the outlet velocity reaches a sufficient magnitude—typically exceeding 180 m/s—to breakdown the rock.

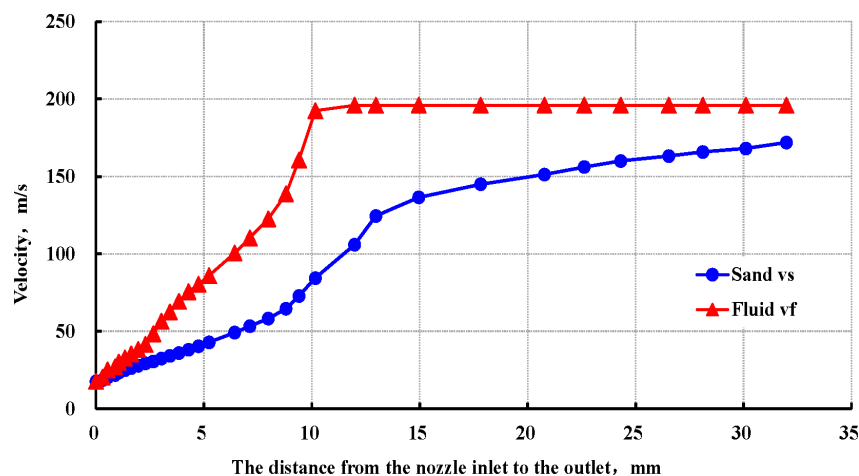


Figure 6—Velocity of distribution of sand and fluid. Calculation parameters: inlet diameter is 8 mm, outlet diameter is 4.56mm, fluid density is 1,000 kg/m³, and surface pump rate is 0.7 m³/min.

After simulation, the recommended jetting parameters as shown in Table 1.

Table 1—Recommended Jetting parameters

No.	Parameters	Volume
1	Sand concentration, kg/m ³	100~120
2	Jetting speed, m/s	>180
3	Based fluid	Linear Gel
4	Silica sand, mesh	40/70
5	Sand volume, m ³	1
6	Pumping rate, L/min	700
7	Perforation fluid, m ³	50

Multi-Stage Procedure

Considering the open hole section with too-free isolation, 2-3/8" CT was recommended to execute the precision multi-stage stimulation. First, running CT with downhole tools to total depth (TD) in a single trip, and conducting the 1st stage stimulation with real-time monitoring of pumping rate and pressure. Once the first stage was completed, immediately pulling back to the 2nd stage, and repeating until the entire horizontal section was precisely stimulated. Three risks needed significant consideration. 1. Borehole instability at toe section. Although well logging revealed relatively uniform physical properties, mud-rich carbonates and chert were identified at the toe during drilling, which pose risks of hole collapse. 2. Sand deposition & CT stuck. Increased leakage at the toe may cause sand deposition accumulation. As little as 0.1 m³ sand can bury 5.5m of CT, making it impossible to pull back. 3. Rapid acid-rock reaction creates deep penetration of wormholes increasing fluid loss by 40-70%, increasing the risk of screen out during sand slurry stage.

Our operation team designed a combined sand jetting & pinpoint acidizing stimulation approach: This involved abrasive jetting and acidizing at every 100m spacing and with another pinpoint acidizing at mid-points between jetting locations (50m offset). The sand jetting parameters were as per Table 1, and the pinpoint acidizing volume was designed to be 50m³, schematic as shown in Fig. 7. The entire horizontal section was divided into 13 jets and 25 pinpoint injections, with a designed sand volume of 13 m³, 390 m³ of linear rubber, and 1,370 m³ of 20% HCL.

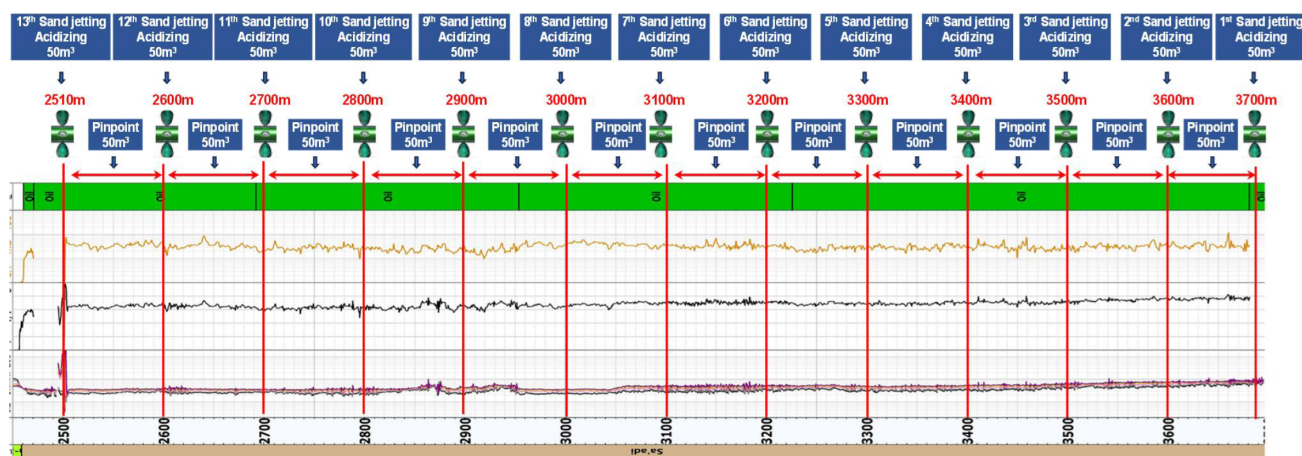


Figure 7—The optimization of sand jetting and pinpoint acidizing

Operation

On March 8, 2025, the acid stimulation operation was executed using optimized parameters adjusted in real-time based on downhole pressure response. The operation was divided into three phases:

Phase I: Toe-Section Acid Jetting (Stages 1-3). Once the 2-3/8" CT coiled tubing reached the bottomhole, to prevent the risk of sand settlement/sticking risks, only 20% hydrochloric acid was pumped. The flow rate was increased from 0.6 to 1.0 m³/min, triggering violent fluctuations in operation pressure and annular pressure, indicating that unstable fluid intake by the formation or failure to develop deep-penetrating wormholes, as shown in Fig. 8. Therefore, Stages 1-3 at the toe were modified to acid-only treatments at 50 m³/stage. This adaptation achieved three critical objectives: evaluating the injectivity of the formation, assessing the reactivity of muddy carbonates and verifying the sand absorption capacity of the formation while eliminating the risk of screenout.

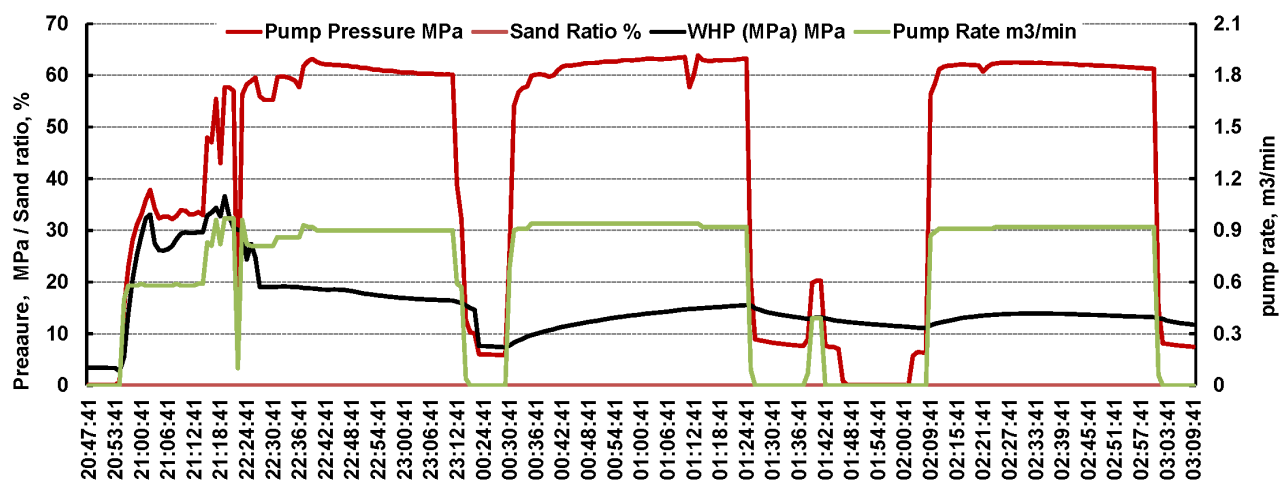


Figure 8—Operation curves of stages 1-3: Pumping pressure and annular pressure (WHP) gradually stabilize.

Phase II: Pseudo-Stable Acid Absorption (Stages 4-6). During the 3rd stage acid treatment, the annular pressure stabilized at 14 MPa (surface pressure + hydrostatic head $\approx$ bottomhole pressure), indicating that a stabilized fluid intake by the reservoir. Field engineers implemented the second-phase adjustment, beginning sand jetting at the 4th stage and maintaining a stable surface pumping rate of 1.0 m³/min and a sand ratio of 7%. Real-time modeling and analysis revealed that when irregular sand particles impacted the reservoir at high speed from the nozzle, the maximum impact velocity reached 180 m/s, resulting in a radial penetration depth of 2-3 m. A sudden increase in leakage caused the surface pressure to drop by approximately 8 MPa, as shown in Fig. 9. This pressure response is still relatively low compared to the typical carbonate responses in this field. Therefore, after the 4th sand jetting, the original acid volume 50 m³/stage was maintained.

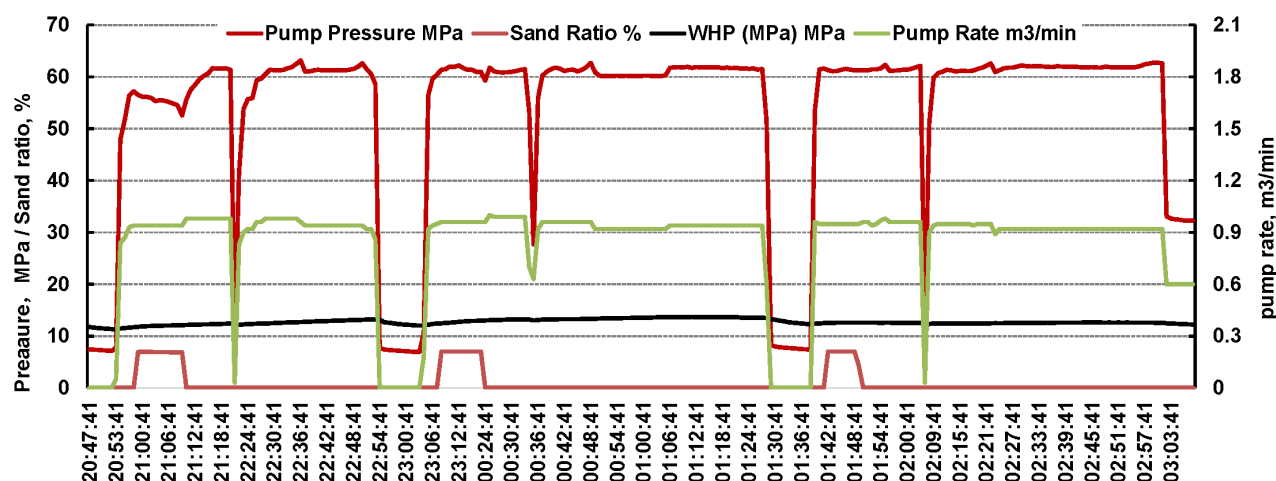


Figure 9—Operation curves of stages 4-6: Pumping pressure decreases slightly, and the annular pressure (WHP) has stabilized

Phase III: Stabilized Jetting & Stimulation (Stages 7-25). After analyzing the operation pressure, engineers found that the reservoir had poor acid absorption capacity and was tighter than expected. Therefore, the acid treatment was adjusted to low-volume sand jetting at each pinpoint with a concentration of 7%, and the acid volume was adjusted within the range of 20 to 50 m³ based on the real-time pressure response. During Stage 7 treatment at 1.0 m³/min (63 MPa pumping pressure, 7% sand concentration), a 12 MPa pressure decline was observed. This pressure drop was also observed at each subsequent stage operation, as shown in Fig. 10. This phenomenon indicates that the sharp increasing in seepage area has led to a rapid decrease in pressure, a fishbone-shaped main fracture has been formed, and acid has leaked laterally from the main channel, forming branching deep-penetrating wormholes. This stage of stimulation proved that the purpose of Fishbone Fracture Acidizing Stimulation had been achieved. On-site engineers reduced the acid volume from 50 m³ to 30 m³ and then to 20 m³, while simultaneously reducing the sand volume from 1.0 m³ to 0.5 m³.

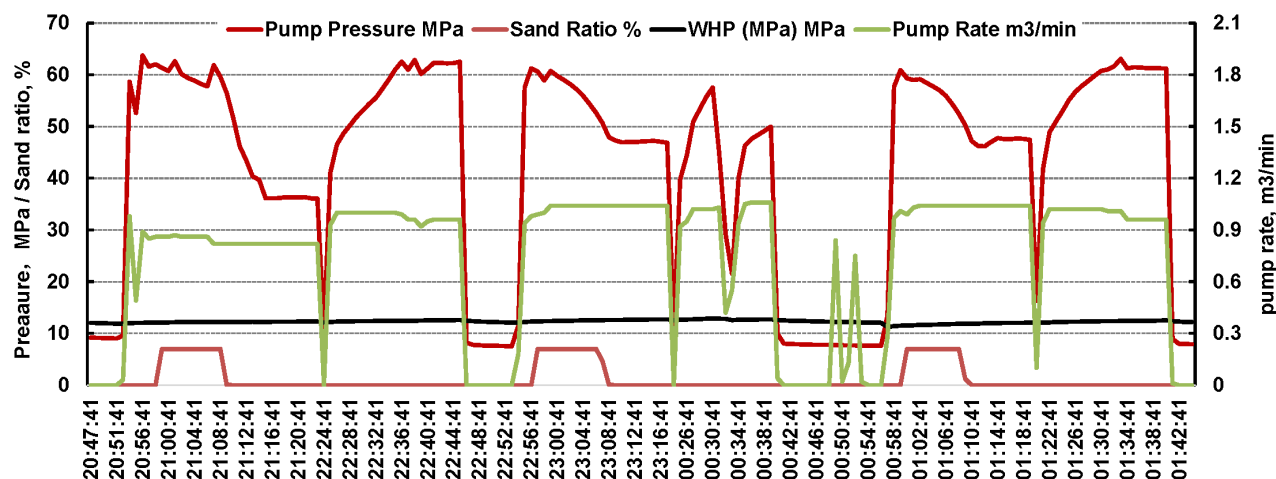


Figure 10—Operation curves of stages 12-14: When the fishbone fractures and wormholes were formed, the pumping pressure drops sharply

Finally, sand jetting was used to establish an infinitely conductive fishbone main fracture (a total of 11 m³ of quartz sand), and acid penetrating was executed to form branching wormholes (a total of 727 m³ of 20% HCL). Through real-time pressure response analysis and dynamic control of operating parameters, a 25-stages tool-free acidizing stimulation of the 1,200 m horizontal section was achieved.

Performance

Post-treatment well testing reached 10,647 bbl/d which is 127 times of the vertical wells production. Diagnostic data confirmed: 25 distinct stimulation zones (Fiber optic logging), 2.2m average fracture penetration (Pressure fitting), and wormhole density increase from 3/cm² to 28/cm². Fiber optic logging revealed that 23 primary fractures with high density wormholes each, creating a fishbone-like infinite conductivity zone.

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